AI-ASSISTED DATA VISUALIZATION FOR BUSINESS DECISION SUPPORT SYSTEMS

An Inte**grated Design, Predictive Analytics, Explainable AI, and Decision-Support Framework**

Research Paper Draft

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ABSTRACT

This paper presents an AI-assisted business analytics and decision-support system that integrates exploratory data analysis, feature engineering, hybrid feature selection, supervised profit prediction, explainable AI, rule-based natural-language question analysis, and interactive visualization into a single workflow. The system is designed around three benchmark business datasets: Tableau Sample Superstore, Global Superstore, and the Brazilian E-Commerce Public Dataset (Olist). Because the datasets differ in schema and target availability, they are analyzed independently rather than being naively merged. Superstore is used as the primary modeling dataset because it contains transaction-level sales and profit fields; Global Superstore and Olist provide complementary global and customer-geographic analytical perspectives. The primary modeling task is transaction-level profit estimation. A hybrid feature-selection stage combines mutual-information ranking with Random Forest importance to select a compact predictor set. Linear Regression and Random Forest are then compared using MAE, RMSE, and R^2. On the held-out Superstore test set, Random Forest achieved MAE 21.10, RMSE 144.13, and R^2 0.7685, compared with MAE 62.30, RMSE 214.96, and R^2 0.4850 for Linear Regression. Permutation-based explainability identified sales and discount as the most influential features for the trained model. A decision-support layer converts these analytical outputs into business summaries, descriptive recommendations, and a rule-based natural-language query interface. The final Streamlit dashboard provides synchronized filters, KPI cards, analytical charts, predictive outputs, explainability, recommendations, and interactive question analysis. The main research contribution is a project-specific hybrid decision-support pipeline that connects feature selection, predictive modeling, model explanation, and actionable analytics rather than presenting visualization and machine learning as separate components. The paper reports measured project results and explicitly distinguishes observed results from future extensions and causal claims.

Keywords: business intelligence; decision support; data visualization; machine learning; Random Forest; feature selection; explainable AI; permutation importance; retail analytics; Streamlit

1. INTRODUCTION

1.1 Background

Business intelligence (BI) has progressed from static reporting toward integrated analytical environments in which data preparation, multidimensional analysis, visualization, and decision support are combined. Chen et al. describe BI and analytics as a broad field connecting data, technologies, applications, and organizational decision making [1]. Data warehousing and OLAP research established the importance of organizing analytical data for efficient multidimensional analysis [2]. More recent research emphasizes that value is not created by data volume alone; users must be able to obtain, interpret, and apply trustworthy information [3].

Interactive dashboards have become a major interface for this process because they compress large analytical spaces into KPIs, charts, tables, filters, and drill-down views. However, dashboard design can introduce information overload and cognitive load. Recent empirical research has shown that dashboard format, information currency, and information completeness can influence decision quality through perceived task complexity and information satisfaction [12]. Other work has explicitly linked dashboard information, representational, and non-information loads with adoption of BI decision-support dashboards [13]. Therefore, an effective business analytics system must address both computational and human-facing concerns.

The present project extends this idea by connecting the dashboard layer to predictive analytics and explainability. Instead of treating visualization as the final step after analysis, the system uses the dashboard as a decision-support surface for observed business performance, profit prediction, feature importance, and business recommendations. The system also includes a natural-language question layer that maps common managerial questions to analytical operations.

1.2 Problem Statement

Retail and e-commerce datasets contain high-dimensional transactional, geographic, product, and customer information. Analysts may need to perform repeated operations such as identifying high-profit categories, comparing regions, understanding discount patterns, predicting profit, and interpreting model outputs. When these operations are performed independently, the analytical workflow becomes fragmented and the connection between data preparation, model output, explanation, and business action can be lost. The central problem addressed in this work is therefore:

How can an integrated business analytics system combine reliable exploratory analysis, predictive modeling, explainability, and interactive decision support so that analytical outputs remain traceable from raw business data to business-facing insights?

1.3 Research Gap

Existing BI literature provides strong foundations for data integration, analytical modeling, dashboards, usability, and decision support [1]-[6]. Machine-learning research separately provides mature techniques for prediction and feature selection [14]-[18], while explainable-AI research provides methods for understanding model behavior [19]-[24]. However, these areas are often presented as separate technical components. A dashboard may show descriptive KPIs without predictive explanation; a predictive model may be evaluated without a business-facing visualization layer; and an XAI method may be reported without connecting explanations to operational recommendations. Recent work on machine-learning-infused dashboards demonstrates the value of connecting ML methods to interactive business analytics [11], motivating an integrated artifact.

The gap addressed here is therefore an artifact-level integration gap: a compact, reproducible pipeline is required that links multi-dataset EDA, hybrid feature selection, predictive profit modeling, model explanation, business-rule recommendations, and natural-language analytical interaction. The contribution is not claimed as a new fundamental ML algorithm; instead, the proposed hybrid pipeline and its explicit connection between model evidence and decision-support functions constitute the project-level methodological novelty.

1.4 Research Objectives

- RO1: Build a reproducible business-data preparation and EDA pipeline across three benchmark retail/e-commerce datasets.
- RO2: Engineer and select informative transaction-level features for profit estimation using a hybrid mutual-information and Random Forest ranking mechanism.
- RO3: Compare a linear baseline and a nonlinear ensemble model for transaction-level profit prediction using MAE, RMSE, and R^2.
- RO4: Apply model-agnostic permutation importance to expose the relative influence of selected predictors.
- RO5: Integrate descriptive analytics, predictive outputs, explanations, recommendations, and natural-language question analysis into an interactive dashboard.
- RO6: Evaluate the resulting system through quantitative model metrics, analytical consistency checks, and reproducible visual evidence.

1.5 Research Questions

- RQ1: What data-preparation and EDA workflow can produce consistent analytical inputs from heterogeneous business datasets?
- RQ2: Does a hybrid feature-selection mechanism provide a compact predictor set for transaction-level profit estimation?
- RQ3: How do Linear Regression and Random Forest compare for profit prediction on the prepared Superstore dataset?
- RQ4: Which selected features are most influential according to permutation importance?
- RQ5: How can predictive and explainable outputs be connected to interactive business decision support?

1.6 Contributions

- A three-dataset business analytics workflow that preserves dataset-specific schemas instead of forcing incompatible sources into a single table.
- A hybrid feature-selection step combining mutual information and Random Forest importance before model training.
- A comparative profit-prediction experiment with a transparent linear baseline and a nonlinear Random Forest model.
- An explainability layer based on permutation importance with both global feature rankings and prediction-level examples.
- A decision-support layer that links descriptive findings, predictive evidence, explainability, business recommendations, and rule-based natural-language questions.
- An interactive Streamlit dashboard that exposes synchronized filters, KPI summaries, charts, predictive analytics, explainability, and recommendations.

2. RELATED WORK

2.1 Business Intelligence and Decision Support

BI is widely positioned as an organizational capability that transforms heterogeneous data into information for decision making [1], [3]. Foundational data-warehouse and OLAP work emphasizes extraction, cleaning, loading, multidimensional structures, and analytical querying [2]. Critical-success research also shows that BI implementation depends on data quality, management support, implementation processes, and organizational readiness [4]. System-success models such as DeLone and McLean emphasize information quality, system quality, use, satisfaction, and net benefits [5]. Technology acceptance research further shows that perceived usefulness and effort-related factors influence technology adoption [6].

2.2 Dashboard Visualization and Human Factors

Dashboard design is closely related to visual analytics and human-computer interaction. Shneiderman's overview-first, zoom-and-filter, details-on-demand principle provides a useful interaction model for analytical interfaces [7]. Munzner's visualization framework emphasizes the relationship among data abstraction, task abstraction, visual encoding, and interaction [8]. Recent dashboard studies provide empirical evidence that visual format, information completeness, and currency can influence decision quality [12], while cognitive-load research shows that excessive dashboard content can hinder adoption [13]. These findings motivate the present system's use of progressive disclosure through navigation, filters, KPI cards, charts, tables, and dedicated analytical sections.

2.3 Machine Learning for Business Prediction

Supervised machine learning is commonly used to estimate continuous business outcomes such as revenue, demand, or profit. Linear Regression provides a transparent baseline, while tree ensembles can model nonlinear interactions. Random Forests combine multiple randomized decision trees and have demonstrated robustness and useful variable-importance mechanisms [14]. Gradient boosting provides another established tree-based strategy for regression and classification [15]. The broader statistical-learning literature provides a theoretical foundation for model selection, regularization, tree methods, and predictive evaluation [16].

2.4 Feature Selection

Feature selection aims to remove irrelevant or redundant variables while improving prediction, efficiency, or interpretability [17]. The present study combines mutual information, which captures general statistical dependence, with model-based Random Forest importance. The resulting combined ranking is used as a compact, project-specific feature-selection mechanism. This is deliberately positioned as a hybrid selection step rather than as a new feature-selection algorithm in the theoretical sense.

2.5 Explainable AI

Explainability is increasingly important when predictive outputs are used to support human decisions. LIME explains individual predictions by learning an interpretable local model around a prediction [19]. SHAP provides a unified additive feature-attribution framework grounded in Shapley values [20]. Broader XAI research distinguishes explanation

methods by their goals, scope, and evaluation assumptions [21]-[24]. In this project, permutation importance is selected because it is model-agnostic and directly compatible with the trained Random Forest pipeline. It measures the change in predictive performance after randomly shuffling a feature and therefore indicates model dependence, not causal influence.

2.6 Dashboard-Integrated Analytics

Recent research has demonstrated the value of dashboards that incorporate machine-learning and analytics methods rather than limiting dashboards to descriptive reporting [11]. A related 2024 experimental study found that visualization format and information quality can affect decision quality through task complexity and information satisfaction [12]. The present work extends this design direction by explicitly placing predictive profit estimation and explainability alongside descriptive business analytics and recommendation logic.

2.7 Comparative Positioning

Research direction	Typical emphasis	Limitation addressed by this project
BI and analytics [1]-[4]	Data integration, analytical value, organizational decision support	Often does not combine prediction and XAI in the same student-scale artifact
Dashboard research [7], [8], [12], [13]	Visualization, interaction, cognitive load	Usually evaluates dashboard interaction rather than end-to-end ML evidence
Machine learning [14]-[16]	Prediction and model performance	Model outputs may remain disconnected from managerial visualization
Feature selection [17]	Relevance and dimensionality reduction	Often studied independently from dashboard decision support
XAI [19]-[24]	Model interpretation and trust	Explanations may not be converted into business-facing recommendations
ML-infused dashboards [11]	Integration of ML and interactive analytics	This project adds a hybrid selection layer, permutation XAI, and natural-language rule routing

3. DATASETS AND DATA PREPARATION

3.1 Dataset Portfolio

The project uses three benchmark business datasets. The datasets are not merged into one modeling table because their schemas, granularities, and target variables differ. Instead, each dataset is cleaned and analyzed independently, while the Superstore dataset is used for the primary profit-prediction experiment.

Dataset	Purpose	Rows / columns used	Modeling role
Tableau Sample Superstore	Retail transaction analytics, profitability, product/category/region analysis	10,194 / 26 after cleaning	Primary ML dataset
Global Superstore	Cross-market and global retail comparison	51,290 / 28 after cleaning	Complementary EDA
Brazilian E-Commerce Public Dataset (Olist) - customers file	Customer geography and repeat-customer profile analysis	99,441 / 6 after cleaning	Complementary EDA

Tableau describes Sample Superstore as a sample business dataset containing product, sales, and profit information for identifying areas of improvement in a fictitious retail company [29]. The Global Superstore source used in this project contains 51,290 order records [30]. The Olist dataset is a real, anonymized Brazilian e-commerce dataset covering approximately 100,000 orders and multiple dimensions such as products, customers, payments, freight, and reviews; the customer table used here contains 99,441 records and 96,096 unique customer identifiers [31].

3.2 Data Cleaning and Quality Assessment

The preprocessing stage standardizes column names, converts date fields, checks missing values, identifies duplicate records, validates categorical and identifier fields, and creates cleaned CSV outputs. For Superstore, shipping duration,

profit margin, order year, order month, and month name are derived. For Global Superstore, shipping duration, profit margin, order-year and order-month fields are derived, while the highly incomplete postal-code field is excluded from analytical modeling. For Olist, customer and customer-unique identifiers are validated and geographic fields are standardized; a ZIP-prefix feature is derived.

Potential outliers are flagged using the interquartile-range rule rather than automatically deleted. This distinction is important because large sales or profit values can represent genuine high-value transactions. The quality workflow therefore treats outlier detection as an analytical review step.

3.3 Exploratory Data Analysis

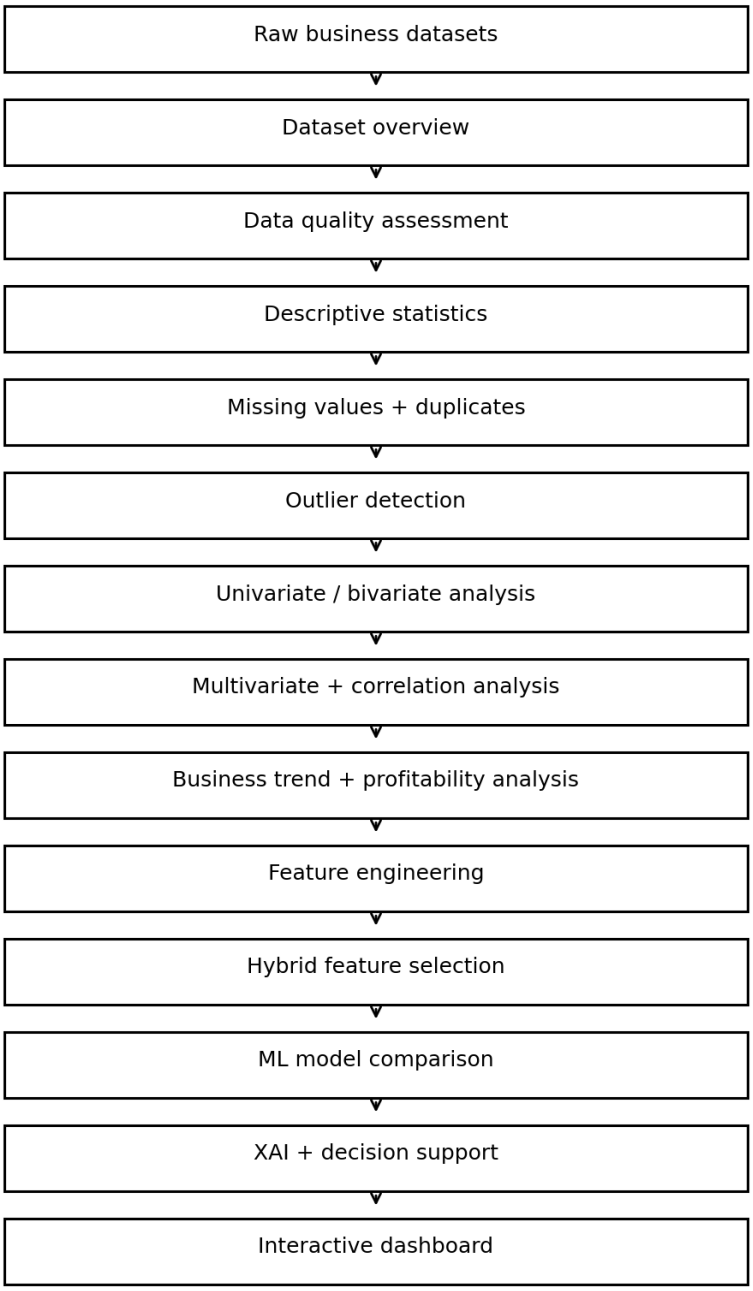


Fig. 2. Project EDA and decision-support workflow.

The EDA process follows the faculty-recommended sequence from raw data and dataset overview through quality assessment, descriptive statistics, missing values, duplicates, outlier detection, univariate, bivariate, multivariate, and correlation analysis, followed by feature engineering, feature selection, and model training [32]. The project extends the

sequence with business-specific sales trends, profitability analysis, cross-dataset geographic analysis, XAI, and dashboard decision support.

Measure	Superstore	Global Superstore	Olist customers
Rows	10,194	51,290	99,441
Columns after cleaning	26	28	6
Sales mean	228.226	246.491	-
Profit mean	28.673	28.611	-
Quantity mean	3.792	3.477	-
Average discount	0.1554	0.1429	-
Notable EDA result	Sales-profit corr. = 0.4815	Sales-shipping-cost corr. = 0.7681	Top state SP = 41,746

3.4 Advanced EDA Findings

For Superstore, IQR screening identified 1,183 potential sales outliers, 1,913 potential profit outliers, 175 quantity outliers, and 874 discount outliers. Correlation analysis produced a sales-profit correlation of 0.4815 and a profit-discount correlation of -0.2189. At category level, Technology had the highest average sales (450.34) and average profit (78.58), while Furniture had average profit of 8.96. These descriptive results are observations and do not establish causality.

For Global Superstore, the sales-shipping-cost correlation was 0.7681, sales-profit correlation was 0.4849, and profit-discount correlation was -0.3165. Market-level analysis showed variation in average sales, profit, quantity, and discount across APAC, Africa, Canada, EMEA, EU, LATAM, and US. For Olist customers, Sao Paulo (SP) accounted for 41,746 records, followed by Rio de Janeiro (RJ) with 12,852 and Minas Gerais (MG) with 11,635. Customer-profile frequency had a mean of 1.035 records and a maximum of 17, indicating that repeat-customer behavior exists in the source.

4. PROPOSED METHODOLOGY AND SYSTEM ARCHITECTURE

4.1 Research Design

The study adopts a design-science orientation because the main research output is an engineered artifact intended to solve a concrete analytical problem. Design science emphasizes building and evaluating artifacts that extend human and organizational capabilities [9]. The DSRM process includes problem identification, objectives, design and development, demonstration, evaluation, and communication [10]. These stages map naturally to the present project: fragmented business analysis is the problem, the integrated AI-assisted analytics system is the artifact, the dashboard is the demonstration, and model/analytical evaluation provides evidence.

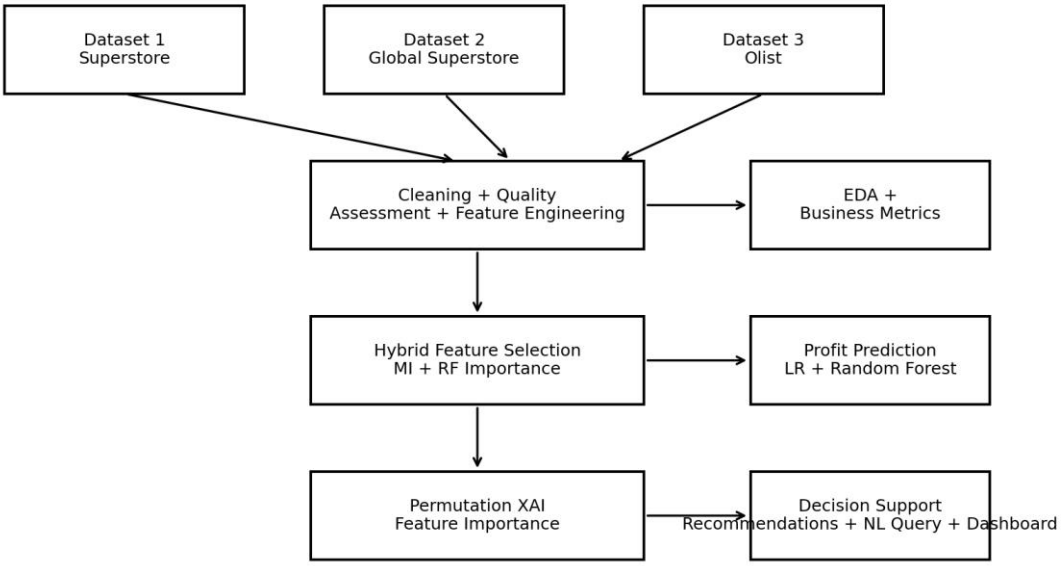


Fig. 1. Proposed AI-assisted business decision-support architecture.

4.2 End-to-End Analytical Pipeline

The system is implemented as a modular Python-based pipeline. The source datasets are cleaned independently. EDA scripts generate descriptive statistics, quality summaries, business metrics, and figures. The Superstore modeling branch then performs feature selection, model training, model comparison, permutation-based explanation, business recommendation generation, and dashboard integration.

Stage	Technique / artifact	Output
Data ingestion	Pandas + dataset-specific loaders	Raw DataFrames
Quality assessment	Missingness, duplicates, type/date validation	Quality report
Feature engineering	Shipping days, profit margin, year/month	Analytical features
EDA	Univariate, bivariate, multivariate, correlation, trend analysis	Statistics + figures
Feature selection	Mutual information + RF importance	Selected 8 features
Prediction	Linear Regression + Random Forest	Profit predictions
Explainability	Permutation importance	Global feature importance
Decision support	Rule-based recommendations + NL query routing	Business insights
Visualization	Streamlit + Plotly	Interactive dashboard

4.3 Feature Engineering

For the Superstore dataset, the derived features include shipping_days = ship_date - order_date, profit_margin = profit / sales where defined, order_year, order_month, and order_month_name. The feature-selection stage excludes the target profit and the derived profit_margin feature from the predictor set to avoid target leakage through a direct transformation of the target.

4.4 Hybrid Feature Selection

The project proposes a hybrid ranking mechanism that combines two complementary signals. First, mutual information estimates the dependency between each candidate feature and profit. Second, Random Forest importance provides a model-based relevance estimate. Both scores are normalized and combined with equal weight to form a project-specific

selection score. The top eight features are retained: sales, discount, subcategory, stateprovince, category, quantity, region, and order_month.

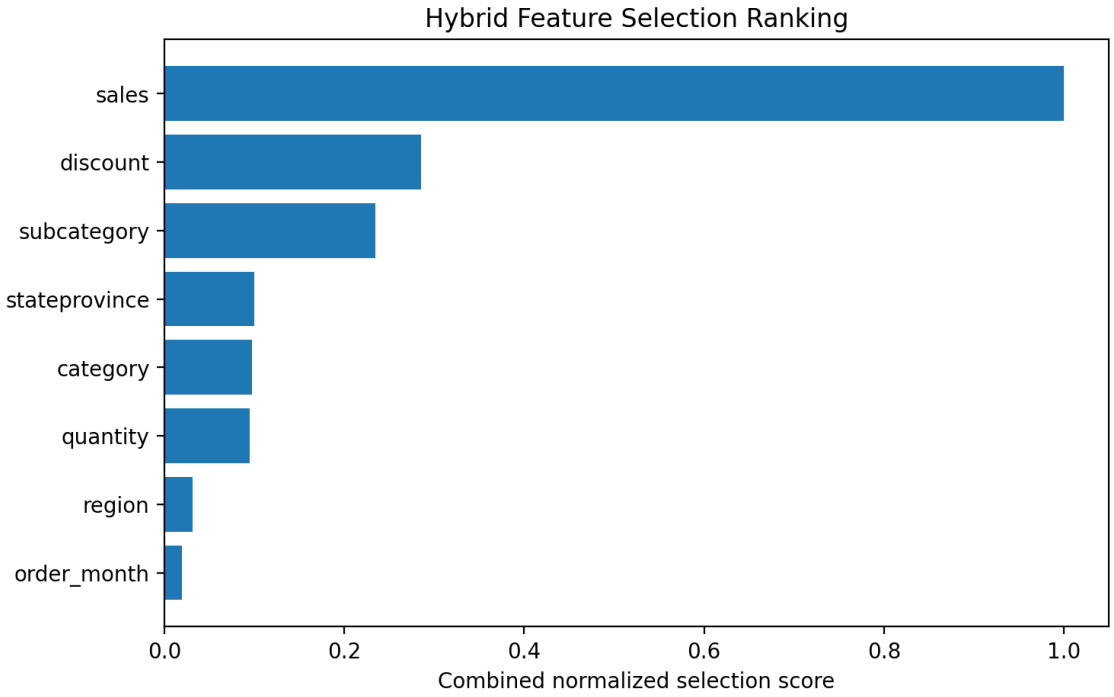


Fig. 4. Hybrid feature-selection ranking used for the profit model.

4.5 Predictive Modeling

Profit is modeled as a continuous target. An 80/20 train-test split produces 8,155 training observations and 2,039 test observations. Numeric features are processed with imputation and categorical features are one-hot encoded in a scikit-learn pipeline [18]. Two models are compared: Linear Regression as a transparent baseline and Random Forest Regressor as a nonlinear ensemble model [14]. The Random Forest configuration uses 200 trees.

Model performance is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and coefficient of determination (R2):

$$MAE = (1/n)\ sum\ |y_i - y_hat_i|$$

$$RMSE = sqrt[(1/n)\ sum\ (y_i - y_hat_i)^2]$$

$$R2 = 1 - [sum(y_i - y_hat_i)^2 / sum(y_i - y_bar)^2]$$

4.6 Explainable AI

Permutation importance is applied to the held-out data using R2 as the scoring function and ten repeated shuffles. For each feature, the model's performance is measured before and after random permutation. A larger decrease indicates greater model dependence on that feature. The resulting importance values are model-specific and should not be interpreted as causal effects.

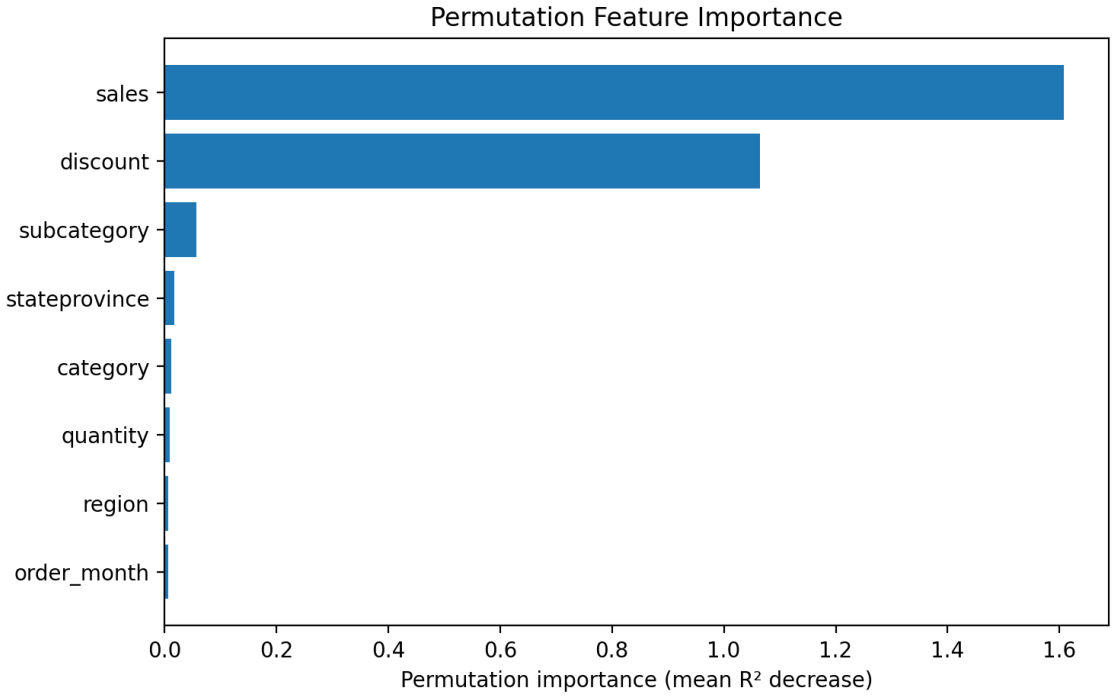


Fig. 5. Permutation importance generated from the trained Random Forest model.

4.7 Decision-Support Layer

The decision-support layer combines four evidence types: descriptive business summaries, model predictions, XAI results, and rule-based business recommendations. The natural-language query component is intentionally rule-based rather than a generative language model. It detects common analytical intent patterns such as category-profit, region-sales, discount-profit, top products, yearly trend, feature importance, and recommendation requests, then calls the appropriate analytical function. This makes the prototype deterministic and auditable.

4.8 Dashboard Design

The Streamlit dashboard provides navigation for Overview, Analytics, Profit Prediction, Explainable AI, Recommendations, and Ask the AI. Global filters include category, region, and order year. KPI cards display total sales, total profit, total quantity, average discount, and average profit margin. Charts and tables respond to the active filter context. The dashboard follows the visual-information principle of overview, filtering, and details on demand [7].

5. EXPERIMENTAL SETUP AND EVALUATION

5.1 Software and Environment

The implementation uses Python with Pandas and NumPy for data processing, scikit-learn for preprocessing, feature selection, model training and evaluation, Matplotlib for analytical figures, and Streamlit/Plotly for the interactive dashboard. The project is organized into dataset-specific analysis modules, feature-selection and modeling modules, XAI analysis, business recommendation generation, decision-support logic, and dashboard code.

5.2 Evaluation Protocol

Evaluation is organized at three levels. First, analytical correctness is checked by comparing generated summaries and figures with independently computed aggregations. Second, predictive performance is evaluated on a held-out test set using MAE, RMSE, and R2. Third, system integration is checked by verifying that dashboard filters, analytical tables, charts, prediction outputs, XAI rankings, and recommendations render successfully. A future publication-grade evaluation should additionally measure usability with a controlled user study, accessibility against WCAG, and web performance metrics such as LCP, INP, and CLS [27], [28].

Evaluation dimension	Metric / test	Evidence in current draft

Data quality	Rows, columns, missingness, duplicates, outlier screening	Dataset-specific preprocessing outputs
EDA correctness	Independent aggregation checks	Category/region/market summaries
Model accuracy	MAE, RMSE, R2	Held-out test-set results
Model comparison	Linear vs Random Forest	Table 5 + Fig. 3
Explainability	Permutation importance, repeated shuffles	Table 6 + Fig. 5
Dashboard integration	Filter and section checks	Dashboard screenshots
Usability	SUS/task study	Not yet conducted; planned
Web performance	LCP/INP/CLS	Not yet instrumented; planned

6. RESULTS AND FINDINGS

6.1 Overall Business Metrics

Metric	Observed value
Total sales	2,326,534.35
Total profit	292,296.81
Total quantity	38,654
Average discount	15.54%
Average profit margin	12.22%

These values are descriptive aggregates of the cleaned Superstore dataset. They are used in the dashboard as the overall business snapshot and change when the user applies global filters.

6.2 Model Comparison

Model	MAE	RMSE	R2
Linear Regression	62.3042	214.9602	0.4850
Random Forest	21.0978	144.1295	0.7685

Random Forest reduced MAE by approximately 66.1% relative to Linear Regression and reduced RMSE by approximately 32.9%. Its R2 was 0.2835 higher. These results indicate that the nonlinear ensemble captured relationships in the prepared transaction features more effectively than the linear baseline for this test split. The result is a predictive comparison, not evidence that Random Forest is universally superior for other business datasets.

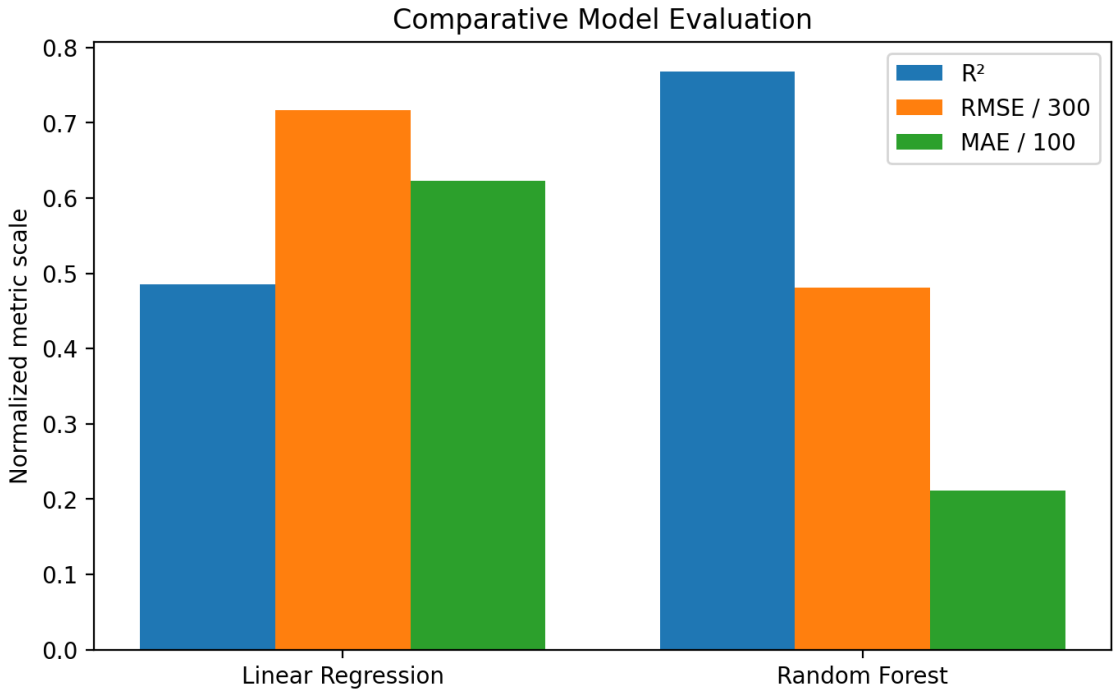


Fig. 3. Comparative model evaluation. RMSE and MAE are scaled in the figure only for visual comparison; exact values are reported in Table 5.

6.3 Feature Selection Results

Rank	Feature	Mutual information	RF importance	Combined score
1	sales	2.053331	0.616181	1.000000
2	discount	0.482385	0.206966	0.285407
3	subcategory	0.775973	0.055656	0.234117
4	stateprovince	0.291728	0.035063	0.099490
5	category	0.361182	0.011613	0.097373
6	quantity	0.356337	0.009874	0.094783
7	region	0.077682	0.015518	0.031508
8	order_month	0.029540	0.015417	0.019704

The selected features are not intended to represent causal drivers of profit. They represent variables that contain predictive information for the trained model. Sales and discount dominate the hybrid ranking, while product and geographic dimensions provide additional context.

6.4 Explainability Results

Rank	Feature	Permutation importance mean	Std.
1	sales	1.607752	0.055588
2	discount	1.064891	0.066204
3	subcategory	0.056778	0.001823
4	stateprovince	0.017471	0.001935
5	category	0.011978	0.001154
6	quantity	0.010081	0.001179
7	region	0.006637	0.001473
8	order_month	0.006349	0.001687

Sales and discount produced substantially larger permutation importance values than the remaining features. The interpretation is that the trained Random Forest's predictive performance depends strongly on these variables in the evaluation data. This is not a causal conclusion. A high permutation importance may arise from correlation, redundancy, or the way the model partitions the feature space.

6.5 Business Analysis Findings

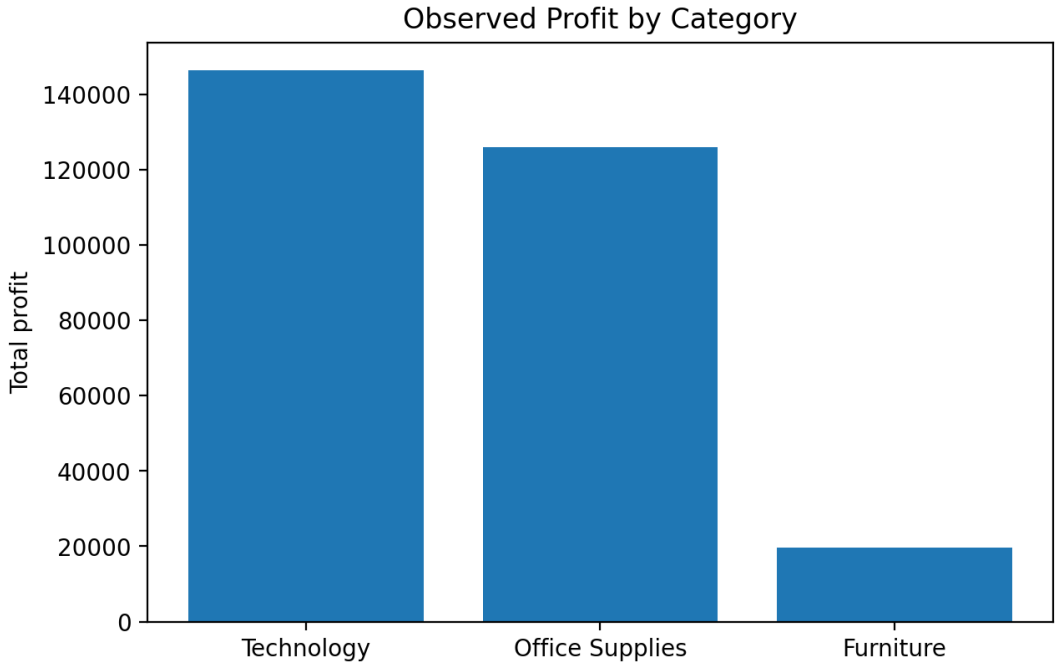


Fig. 6. Observed total profit by Superstore category.

Category	Sales	Profit	Quantity	Average discount
Technology	839,893.28	146,543.38	7,017	0.1315
Office Supplies	731,893.31	126,023.44	23,268	0.1563
Furniture	754,747.76	19,730.00	8,369	0.1730

Technology has the largest observed total profit in the cleaned dataset, while Furniture has the smallest total profit. The dashboard therefore exposes these differences for investigation. The system does not automatically interpret a category as good or bad; recommendations are phrased as areas for analysis, such as reviewing pricing, discounts, product mix, and operating costs.

At regional level, West has observed profit of 110,798.82, followed by East at 94,883.26, South at 46,749.43, and Central at 39,865.31. At subcategory level, Tables, Bookcases, and Supplies have negative total profit and are therefore flagged for review. Discount analysis shows that the 40-60% range has the lowest average profit per transaction at approximately -129.04. This association should not be interpreted as proof that discounts cause losses; transaction mix and product characteristics may confound the relationship.

6.6 Dashboard Evidence

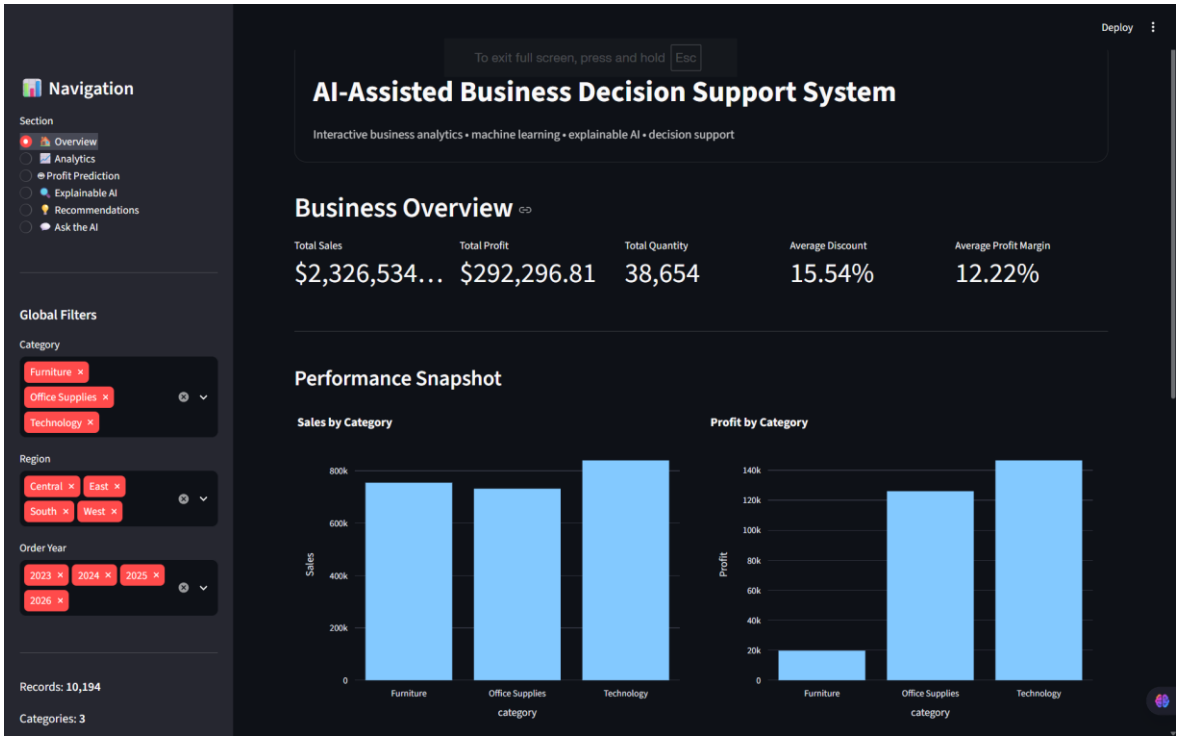


Fig. 7. Dashboard overview with synchronized category, region, and order-year filters and KPI/performance views.

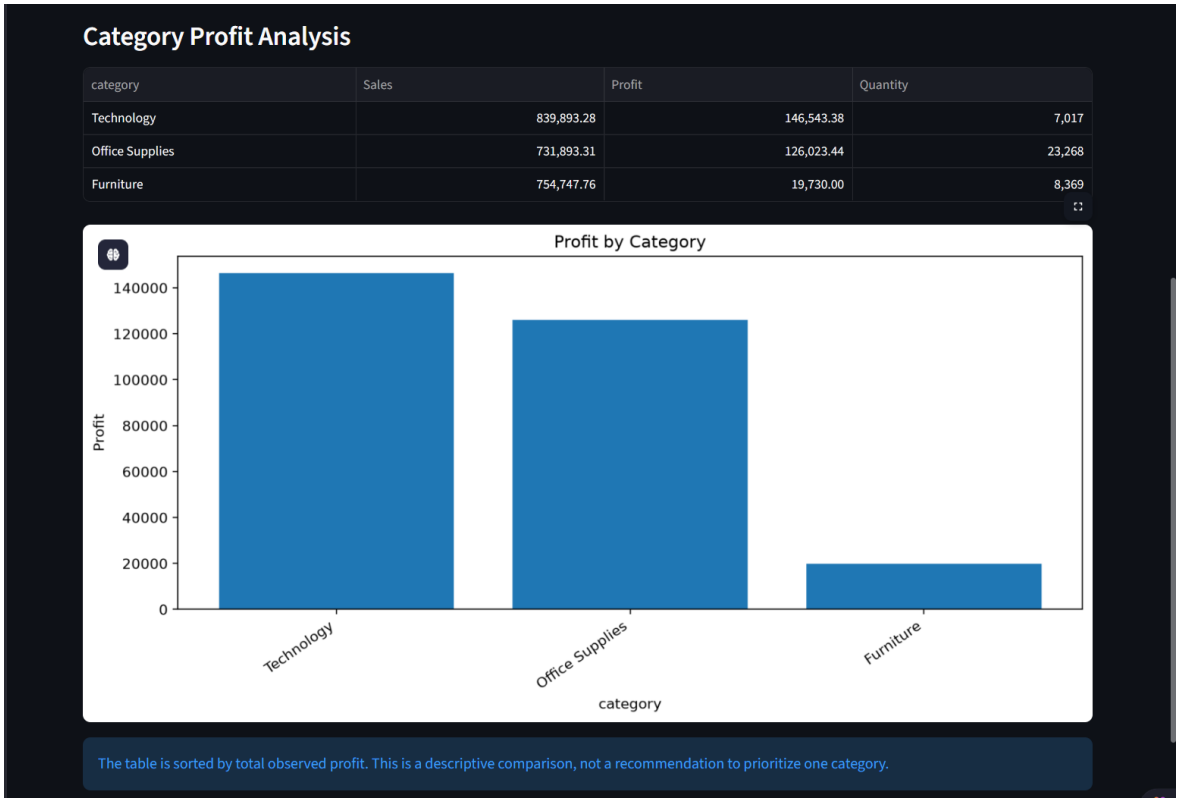


Fig. 8. Dashboard category-profit analysis showing table and chart views.



rank	feature	importance_mean	importance_std
1	sales	1.607752	0.055588
2	discount	1.064891	0.066204
3	subcategory	0.056778	0.001823
4	stateprovince	0.017471	0.001935
5	category	0.011978	0.001154
6	quantity	0.010081	0.001179
7	region	0.006637	0.001473
8	order_month	0.006349	0.001687

Fig. 9. Dashboard explainable-AI view showing permutation feature importance.

The dashboard screenshots provide direct artifact-level evidence of the implemented interface. The overview screen exposes KPI cards and synchronized filters; the category screen links tabular values with a profit chart; and the XAI screen exposes the permutation-importance table. Additional screenshots for Profit Prediction, Recommendations, and Ask the AI should be inserted in the final submission package to satisfy the faculty requirement for complete front-end evidence.

7. COMPARATIVE ANALYSIS AND DISCUSSION

7.1 Comparison with Existing Modeling Approaches

The experimental comparison establishes a transparent baseline-versus-ensemble relationship. Linear Regression provides a low-complexity reference, while Random Forest provides a nonlinear ensemble alternative. The difference in test metrics demonstrates that the selected transaction features contain nonlinear predictive structure that is not fully captured by the linear model. This comparison is stronger than reporting a single model because it provides an explicit benchmark against an established alternative.

7.2 Comparison with Conventional Dashboarding

A conventional BI dashboard can summarize sales, profit, quantity, categories, regions, and trends. The proposed system extends this pattern by adding a predictive model, an explicit explainability layer, and a natural-language rule-routing interface. Consequently, the dashboard is not only a descriptive reporting surface; it also exposes the evidence chain from selected features to predictions and model explanations. This design is aligned with research showing that machine-learning-infused dashboards can make advanced analytics accessible to business users [11].

7.3 Project-Level Novelty

The faculty requirement emphasizes that novelty cannot be claimed merely from building a front-end dashboard or from using three datasets. The novelty claimed in this paper therefore resides in the integrated analytical step added between EDA and decision support: a hybrid feature-selection mechanism combines mutual information and Random Forest relevance, feeds a comparative prediction stage, and passes the selected model into permutation-based XAI. The resulting evidence is then connected to deterministic business recommendations and natural-language question routing. This creates a repeatable evidence chain: data -> hybrid selection -> prediction -> explanation -> decision support -> visualization. The contribution is best understood as a project-specific methodological integration rather than a claim of inventing a new learning algorithm.

7.4 Decision-Support Interpretation

The system intentionally separates observation from recommendation. A category with high observed profit is reported as a descriptive finding; a recommendation may then suggest examining its successful products or pricing patterns. Similarly, a loss-making subcategory is flagged for investigation rather than automatically recommending that it be removed. This distinction reduces the risk of turning descriptive statistics or model importance into unsupported causal or managerial claims.

8. LIMITATIONS AND THREATS TO VALIDITY

8.1 Dataset Scope

The primary ML model is trained on Superstore. Global Superstore and Olist are used for complementary EDA rather than pooled modeling because their schemas and targets differ. Therefore, model performance should not be generalized to all three datasets.

8.2 Prediction Semantics

The model estimates transaction-level profit using sales and other transaction attributes. It is not a future sales forecast. Because sales is a predictor, the model is most appropriate for analytical profit estimation after or during transaction formation rather than pre-sale demand planning.

8.3 Model Generalization

The reported metrics are based on one train-test split. Cross-validation, time-aware validation, and external validation would provide stronger evidence for deployment.

8.4 Explainability

Permutation importance indicates model dependence, not causality. Correlated features may share importance, and the results depend on the evaluation dataset and scoring metric.

8.5 Natural-Language Layer

The question interface is rule-based. It does not constitute a general-purpose conversational AI system and can only answer intents explicitly encoded in the routing logic.

8.6 Human Evaluation

No controlled user study or SUS measurement has been conducted in the current project. Such evaluation is required before making publication-grade claims about usability or decision quality.

8.7 Novelty Boundary

The project does not claim a new Random Forest algorithm, a new XAI theory, or a universally optimal dashboard. The novelty is the implemented hybrid decision-support pipeline and its integration of established techniques.

9. CONCLUSION AND FUTURE WORK

This paper presented an AI-assisted business decision-support system that integrates business-data preparation, exploratory analysis, hybrid feature selection, profit prediction, explainable AI, business recommendations, natural-language rule routing, and interactive visualization. The three-dataset design broadens the analytical scope while preserving dataset-specific validity. The Superstore dataset served as the primary predictive modeling source because it provides transaction-level profit, while Global Superstore and Olist provided complementary global and customer-geographic analyses.

The quantitative results demonstrate a clear difference between the tested models: Random Forest achieved MAE 21.10, RMSE 144.13, and R2 0.7685, compared with MAE 62.30, RMSE 214.96, and R2 0.4850 for Linear Regression. The hybrid feature-selection stage identified sales, discount, subcategory, stateprovince, category, quantity, region, and order_month as the selected predictor set. Permutation importance further showed strong model dependence on sales and discount. The dashboard operationalizes these outputs through KPI cards, synchronized filters, charts, prediction views, explainability, recommendations, and analytical questions.

Future work should strengthen empirical validation through cross-validation or time-aware splits, additional tree-ensemble baselines such as Gradient Boosting or XGBoost where available, calibration of prediction uncertainty, SHAP/LIME comparison, formal user studies, WCAG accessibility testing, web-performance instrumentation, and controlled decision-task experiments. A future version could also incorporate true conversational language understanding, but such an extension should be evaluated separately from the current deterministic prototype.

10. REPRODUCIBILITY AND RESEARCH ARTIFACT

The project maintains a modular structure separating datasets, source scripts, cleaned outputs, model artifacts, figures, dashboard code, and requirements. The core analytical modules include stage1_analysis.py, dataset2_analysis.py, dataset3_analysis.py, advanced_eda.py, feature_selection.py, ml_model.py, xai_analysis.py, business_recommendations.py, ai_decision_support.py, and dashboard/app.py. The final submission should archive the software version, dataset provenance, environment specification, model metrics, generated figures, dashboard screenshots, and the final PPT.

Artifact	Evidence
Cleaned datasets	superstore_cleaned.csv; global_superstore_cleaned.csv; olist_customers_cleaned.csv
Feature selection	feature_selection_results.csv; selected_features.csv; Fig. 4
Model comparison	model_comparison.csv; profit_predictions.csv; Fig. 3
Model artifact	best_profit_prediction_model.pkl
XAI	xai_feature_importance.csv; xai_prediction_explanations.csv; Fig. 5 and Fig. 9
Business recommendations	business_recommendations.csv and summary tables
Dashboard	Streamlit app with Overview, Analytics, Profit Prediction, XAI, Recommendations, and Ask the AI

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APPENDIX A. FACULTY-SUBMISSION EVIDENCE CHECKLIST

- Complete EDA flow and descriptions: included.
- Dataset descriptions for all three datasets: included.
- Detailed methodology and system architecture: included.
- Novelty/new step: hybrid mutual-information + Random Forest feature-selection stage and integrated evidence chain: included.
- Performance comparison using existing methods: Linear Regression vs Random Forest: included.
- Results tables and graphs: included.
- Explainability results and explanation: included.
- Front-end dashboard screenshots: overview, category analysis, and XAI included; additional prediction/recommendation/Ask-the-AI screenshots should be inserted before final submission.
- 30+ references: included (32 references).
- Limitations and evidence boundary: included.
- PPT: to be prepared next from this finalized paper structure.